IBM APPLIED DATA SCIENCE CAPSTONE

SpaceX Falcon 9 First Stage Landing Prediction

Data Science Capstone Project Report — predicting whether the Falcon 9 first stage will land successfully, so the cost of a launch can be estimated.

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Project repository: **https://github.com/SadmanJarif/applied-data-science-capstone**

Executive Summary

Methods. Collected 90 Falcon 9 launches from the SpaceX REST API and Wikipedia web scraping; wrangled the data (cleaning, one-hot encoding, a Class label for landing success); explored it with data visualization and SQL; mapped launch sites with Folium and built a Plotly Dash dashboard; trained four machine-learning models (Logistic Regression, SVM, Decision Tree, KNN) on an 80/20 train–test split (18 test records).

66.7%
Overall first-stage landing success rate (60 of 90 launches)

90.0%
Success rate in 2019 — up from 0% in 2010–2013

77.3%
Best launch site: KSC LC 39A (17 of 22 successful)

94.44%
Best model test accuracy: Decision Tree (17 of 18 correct)

Key results. Landing success improves sharply with flight experience and over time; VLEO orbits succeed most often (85.7%) and GTO least (51.9%) among major orbits; KSC LC 39A is the most reliable launch site. The Decision Tree predicts landing outcome with 94.44% test accuracy, so SpaceX can estimate launch cost from landing success with high confidence.

Introduction

Background

SpaceX advertises a Falcon 9 rocket launch at $62 million, while other providers charge upwards of $165 million. Most of this saving comes from reusing the first stage: SpaceX lands it (on a drone ship or a ground pad) and flies it again instead of losing it in the ocean.

The problem

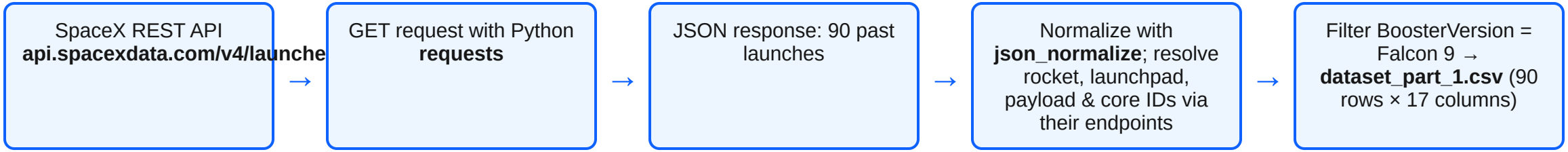
If the first stage cannot be recovered, the launch costs far more. So the key business question is: **can we predict whether the Falcon 9 first stage will land successfully?** If we can predict the landing outcome, we can estimate the cost of a launch.

What this project does

- Collects Falcon 9 launch data from the SpaceX API and Wikipedia (90 launches, 2010–2020).
- Explores which factors affect landing success: launch site, orbit, payload mass, flight number and landing outcome.
- Builds an interactive dashboard and map of launch sites and outcomes.
- Trains machine-learning models to predict landing success (Class = 1) vs. failure (Class = 0).

Data Collection — SpaceX API

Launch data was collected from the **SpaceX REST API** using Python **requests** and **JSON** handling.

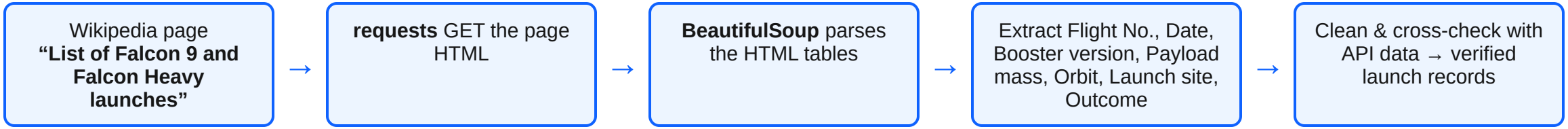


Key phrases: REST API • GET request • JSON • normalization • endpoints (/rockets, /launchpads, /payloads, /cores) • filtering

Fields collected: FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, Latitude. First launch in the data: 2006-03-17.

Data Collection — Web Scraping

Launch records were also scraped from Wikipedia to cross-check and complete the API data.



Key phrases: web scraping • BeautifulSoup • HTML table parsing • requests • data extraction • cross-checking

The scraped table confirms the API data: 90 Falcon 9 launches, all with BoosterVersion = Falcon 9, and the same launch sites, orbits and outcomes used in the analysis.

Data Wrangling Methodology

- **Missing values:** only **LandingPad** has missing values (26 of 90); all other columns are complete. PayloadMass mean is 6,105 kg (min 350 kg, max 15,600 kg).
- **Outcome labels:** the Outcome column (e.g. True ASDS, True RTLS, None None) was converted into a binary target **Class** — 1 = first stage landed successfully, 0 = otherwise. Overall success rate: 66.7% (60 of 90).
- **One-hot encoding:** Orbit (11 categories), LaunchSite (3), LandingPad (5) and Serial (53) were one-hot encoded, giving **80 feature columns** for modelling.
- **Result: dataset_part_2.csv** — 90 rows × 18 columns (dataset_part_1.csv + Class).

Outcome	True ASDS 41	None None 19	True RTLS 14	False ASDS 6
Outcome	True Ocean 5	False Ocean 2	None ASDS 2	False RTLS 1

Code, notebooks and data: **https://github.com/SadmanJarif/applied-data-science-capstone**

EDA with Data Visualization

Charts were drawn with Matplotlib/Seaborn (and Plotly for the dashboard) to see how landing success depends on flight experience, payload, launch site and orbit.

Chart	Purpose
Scatter: FlightNumber vs LaunchSite	Does success improve with flight experience at each site?
Scatter: PayloadMass vs LaunchSite	Does payload mass affect landing success at each site?
Scatter: FlightNumber vs Orbit	Does success improve with experience for each orbit?
Scatter: PayloadMass vs Orbit	Does payload mass affect success for each orbit?
Bar: success rate by orbit	Compare landing success across orbits
Line: yearly success rate	Show the trend in landing success over 2010–2020
Pie: successful landings by site	Share of successful landings from each launch site

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EDA with SQL

The SpaceX launch data (**spacex.csv**, 101 rows) was loaded into a SQLite database (table **SPACEXTBL**) and queried with SQL.

Task	SQL query (summary)
Preview the data	SELECT * FROM SPACEXTBL LIMIT 20
Unique launch sites	SELECT DISTINCT Launch_Site FROM SPACEXTBL
Total payload for NASA (CRS)	SELECT SUM(PAYLOAD_MASS__KG_) WHERE Customer = 'NASA (CRS)'
Average payload, F9 v1.1	SELECT AVG(PAYLOAD_MASS__KG_) WHERE Booster_Version = 'F9 v1.1'
First successful ground-pad landing	SELECT MIN(Date) WHERE Landing_Outcome = 'Success (ground pad)'
Drone-ship successes, payload 4,000–6,000 kg	SELECT Booster_Version WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000
Mission outcome counts	SELECT Mission_Outcome, COUNT(*) GROUP BY Mission_Outcome
Launches by site	SELECT Launch_Site, COUNT(*) GROUP BY Launch_Site

Code, notebooks and data: **https://github.com/SadmanJarif/applied-data-science-capstone**

Interactive Visual Analytics

Folium map

- Markers for all launch sites (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E) and a circle marker for every launch record, coloured green for a successful landing and red for a failure.
- MousePosition and distance measurement, plus a proximity analysis (distance from CCAFS SLC-40 to the coastline, railway, highway and nearest city).

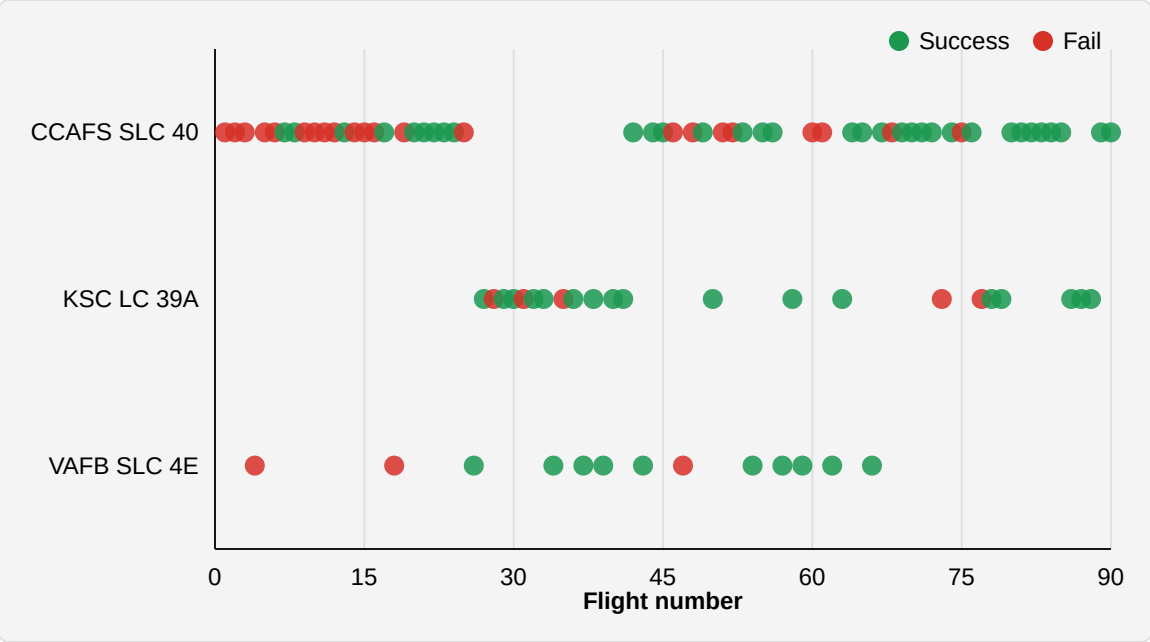
Plotly Dash dashboard

- A dropdown selects a launch site (All Sites, or one site); a pie chart shows successful launches for the selection.
- A payload range slider (0–10,000 kg) filters a scatter chart of payload mass vs. launch outcome, coloured by booster version.

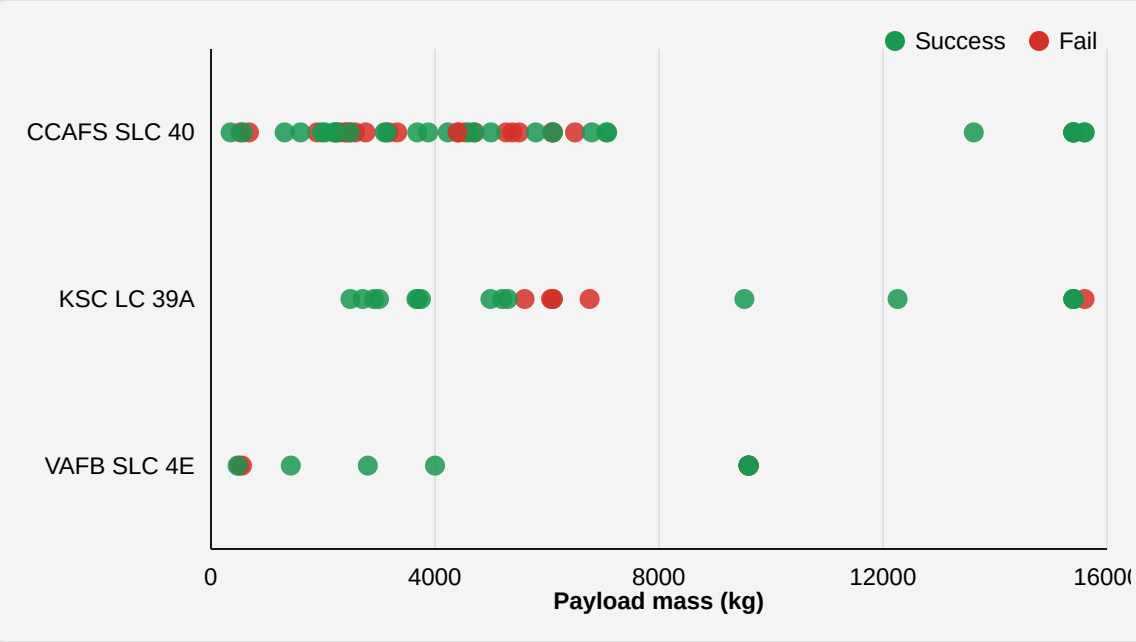
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Flight Number & Payload vs. Launch Site

Flight Number vs. Launch Site



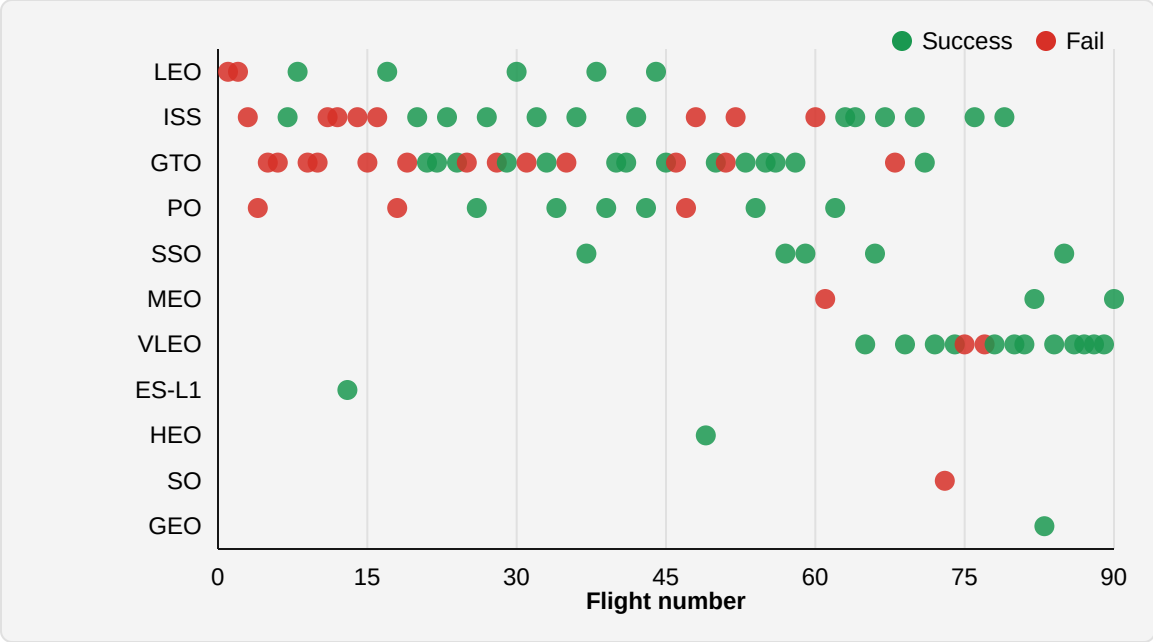
Payload Mass vs. Launch Site



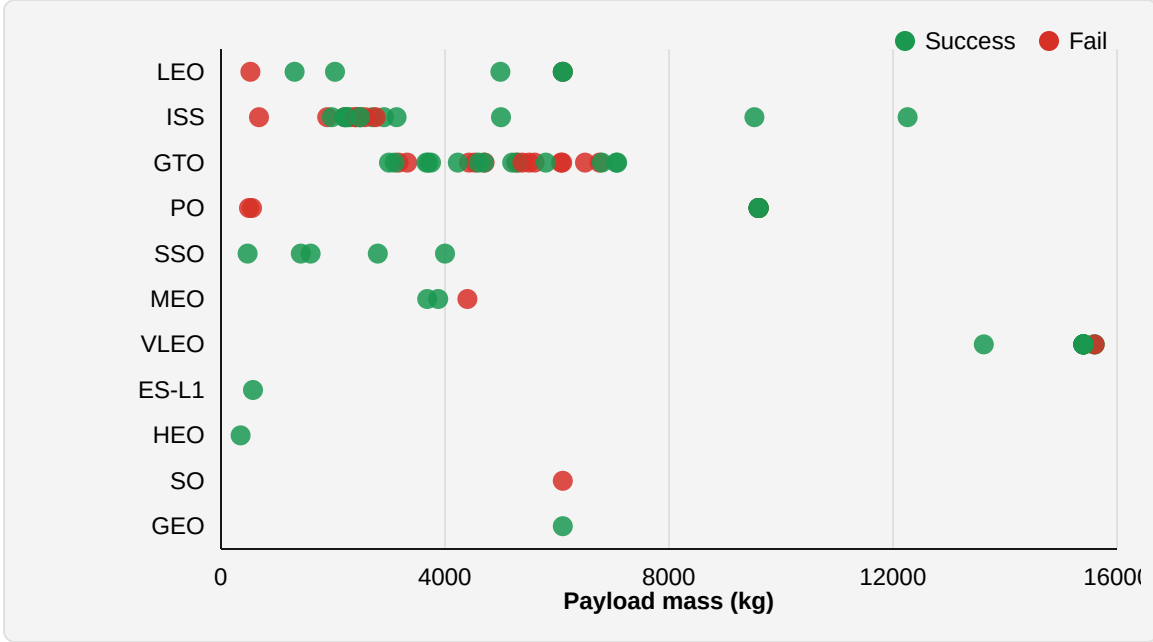
Insight: Landing success rises with flight number — later flights (more experience) land successfully far more often, at every site. Payload mass shows no strong effect on outcome: successes and failures occur across the whole 350–15,600 kg range. CCAFS SLC 40 has the most launches (55) but the lowest success rate (60.0%); KSC LC 39A (77.3%) and VAFB SLC 4E (76.9%) are more reliable.

Flight Number & Payload vs. Orbit

Flight Number vs. Orbit



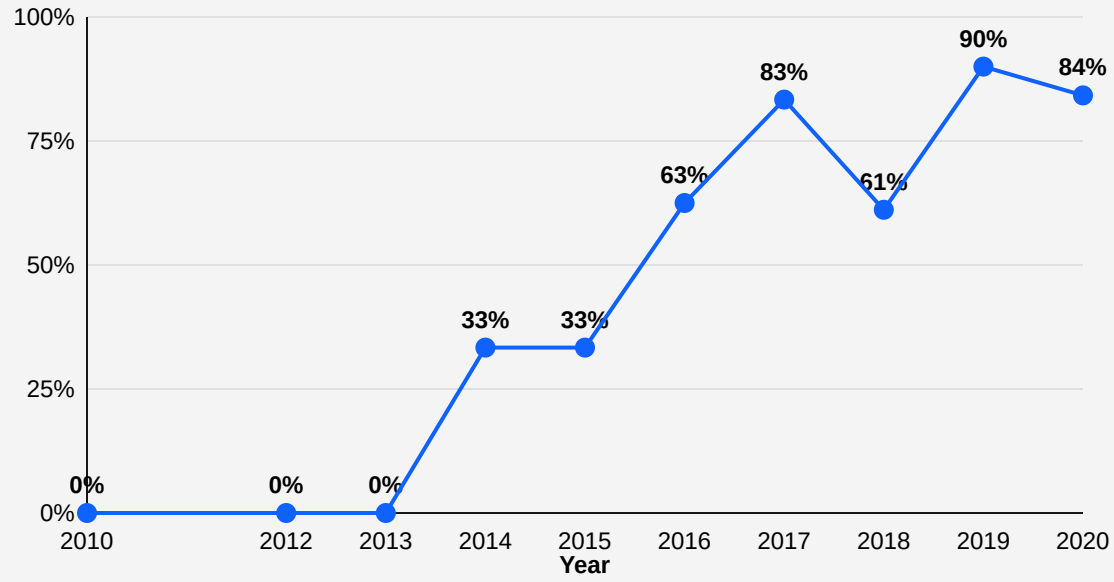
Payload Mass vs. Orbit



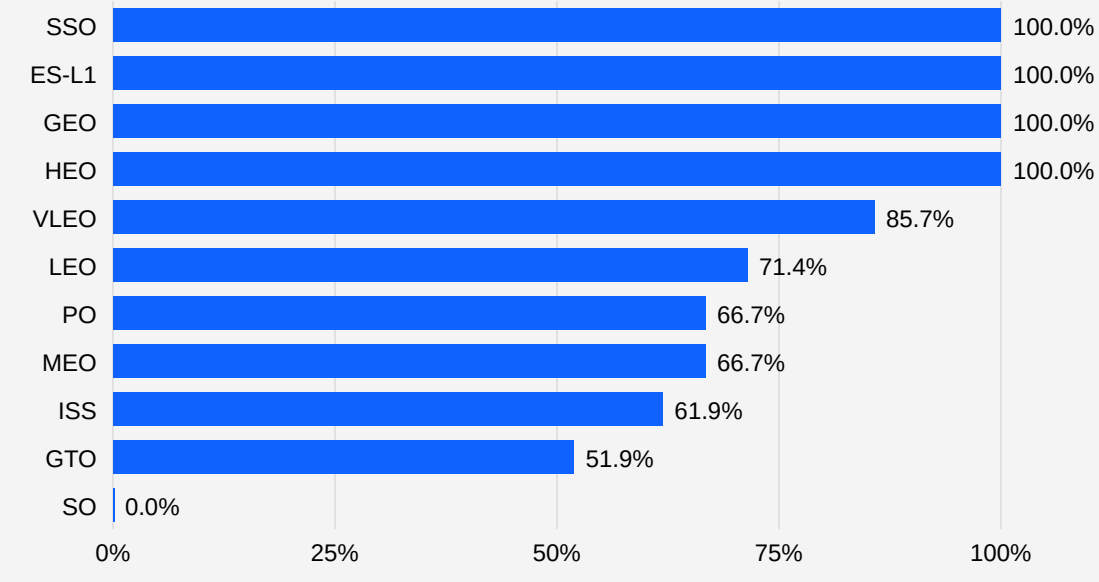
Insight: The same experience effect appears for every orbit: early flights mostly fail, later flights mostly succeed. Heavy payloads (up to 15,600 kg) still land successfully — VLEO missions carry the heaviest payloads and succeed 85.7% of the time. GTO is the hardest major orbit (51.9% success); SSO succeeds 100% (5 of 5).

Success Rate by Orbit & by Year

Yearly landing success rate



Success rate by orbit



Insight: Landing success rose from **0% in 2010–2013** to 33.3% (2014–2015), 62.5% (2016), 83.3% (2017), 61.1% (2018), **90.0% (2019)** and 84.2% (2020). By orbit, SSO (100%, 5 of 5) and VLEO (85.7%, 12 of 14) are the most reliable; GTO is the hardest major orbit at 51.9% (14 of 27); LEO 71.4%, ISS 61.9%, PO 66.7%, MEO 66.7%.

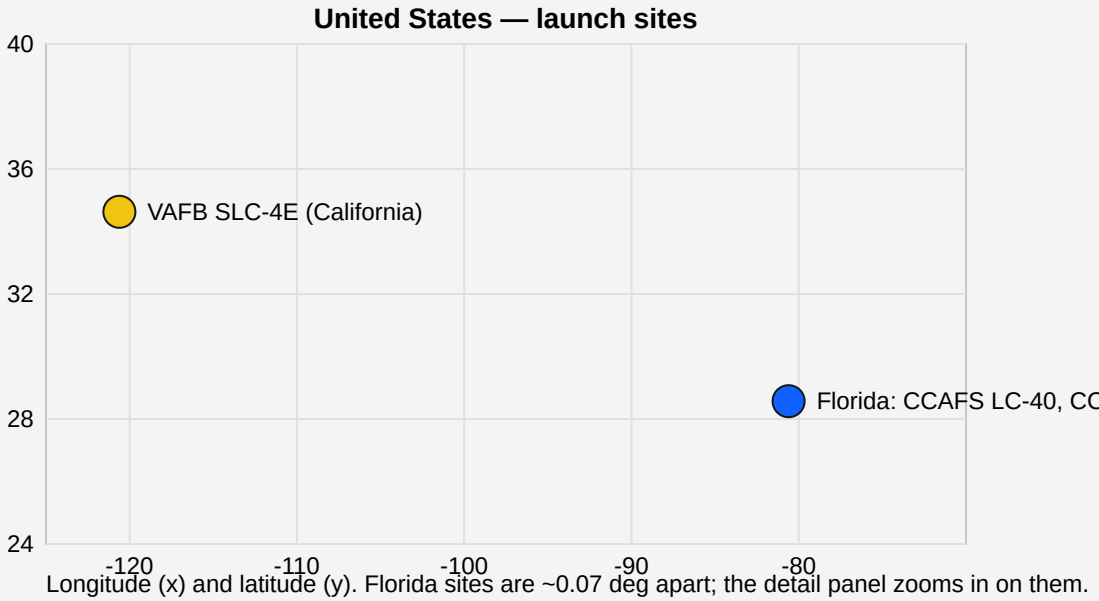
EDA with SQL — Results

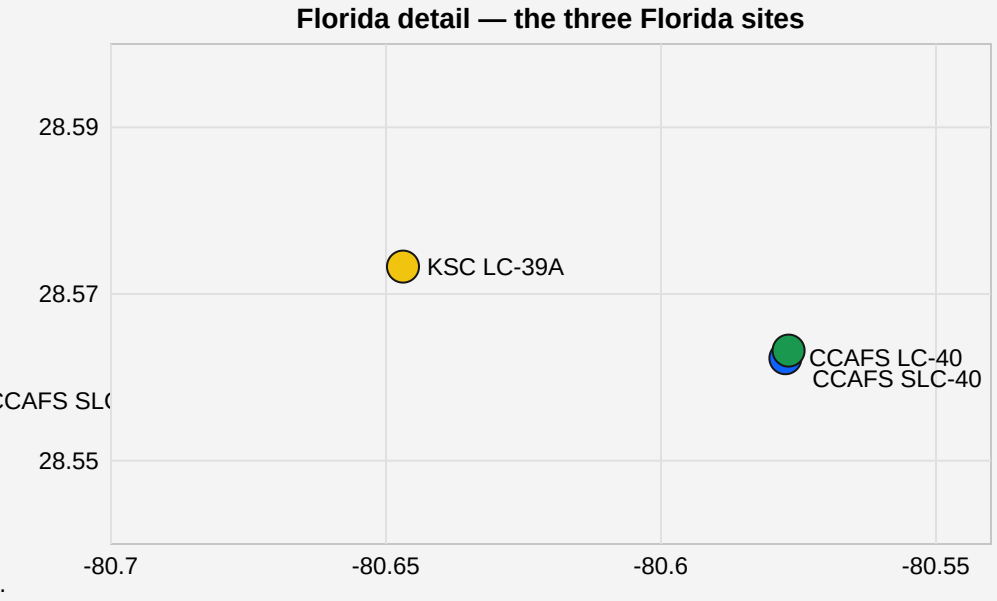
Question	Result (spacex.csv, 101 rows)
Unique launch sites	CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
Total payload mass, NASA (CRS)	45,596 kg
Average payload mass, F9 v1.1	2,928.4 kg
First successful ground-pad landing	2015-12-22
Successful drone-ship landings, payload 4,000–6,000 kg	F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2
Mission outcomes	Success: 100 | Failure (in flight): 1

Launch sites in the 90-launch Falcon 9 dataset

Launch site	Launches	Successful landings	Success rate
CCAFS SLC 40	55	33	60.0%
KSC LC 39A	22	17	77.3%
VAFB SLC 4E	13	10	76.9%

Folium Map — Launch Sites & Launch Records





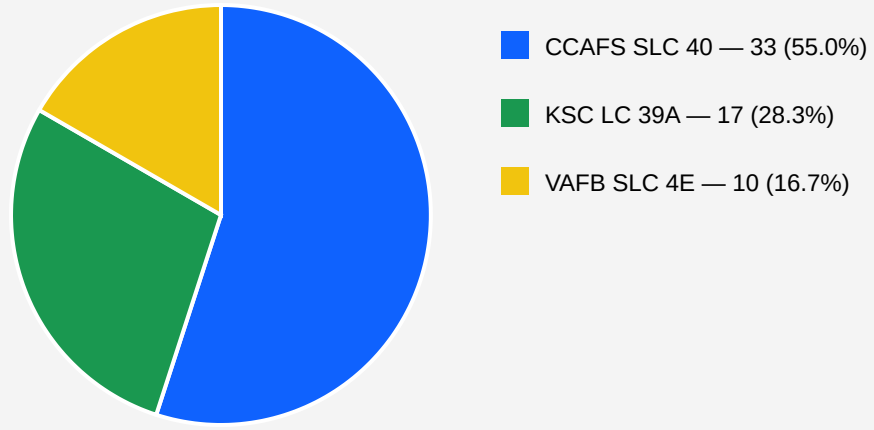
Longitude (x) and latitude (y). Florida sites are ~0.07 deg apart; the detail panel zooms in on them.

Launch site	Latitude	Longitude	Launches (90-launch dataset)
CCAFS LC-40	28.5623	-80.5774	— (in SQL dataset)
CCAFS SLC-40	28.5632	-80.5768	55
KSC LC-39A	28.5733	-80.6469	22
VAFB SLC-4E	34.6328	-120.6107	13

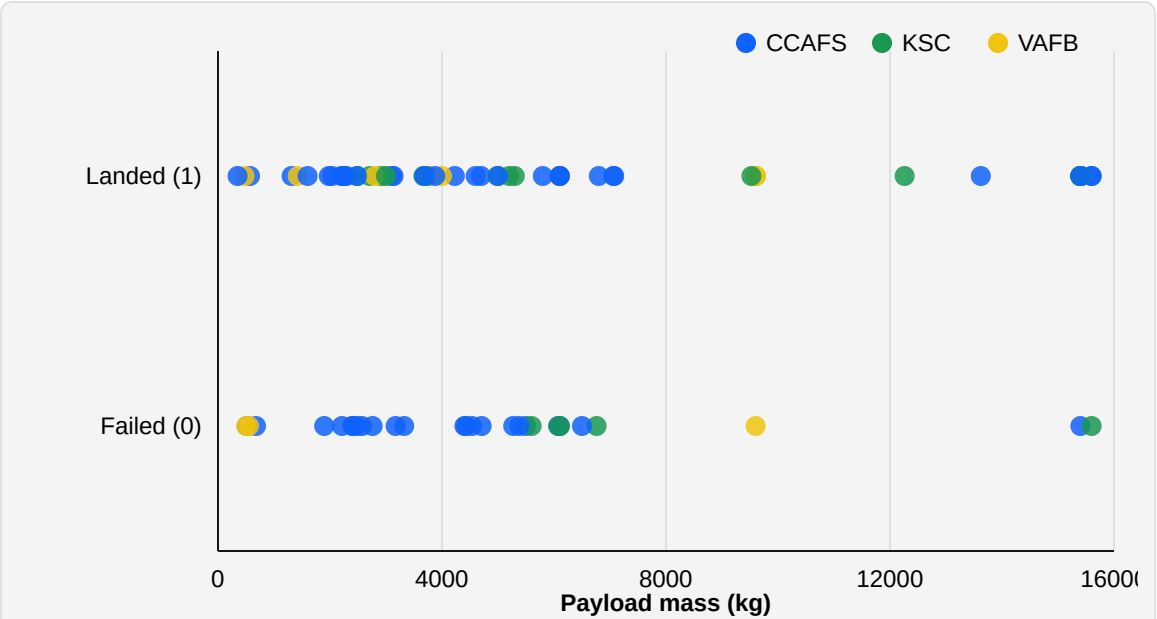
Proximity analysis (CCAFS SLC-40): coastline 0.86 km | railway 1.28 km | highway 0.58 km | nearest city 51.43 km. **Insight:** launch sites sit on coasts (safety), close to rail and highway transport, and far from cities; success markers (green) cluster at KSC LC-39A and VAFB SLC-4E.

Plotly Dash Dashboard — Success & Payload

Successful landings by launch site



Payload mass vs. launch outcome

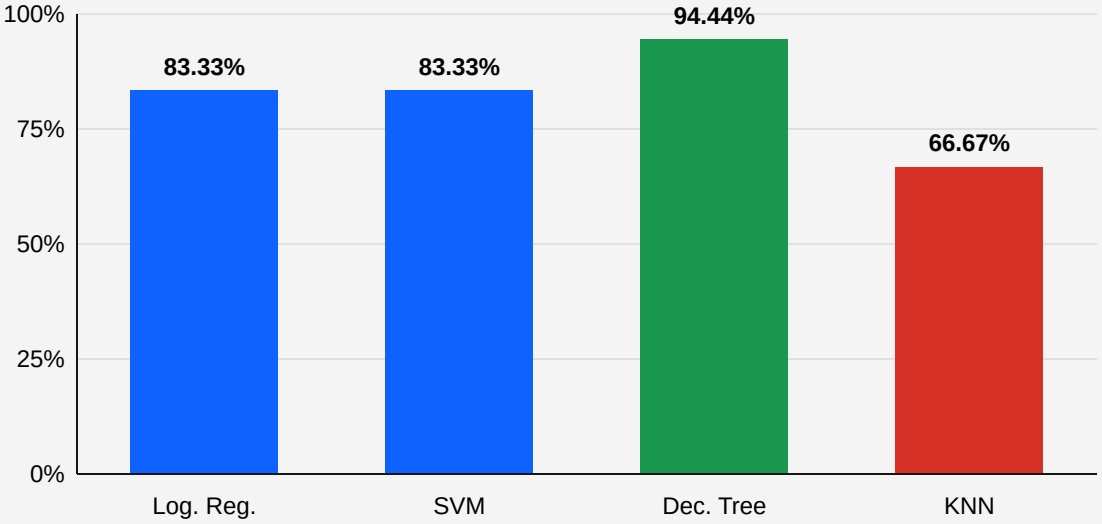


Insight: The dashboard's pie chart shows that CCAFS SLC 40 contributes the most successful landings (33 of 60), while KSC LC 39A has the highest success rate (77.3%). The payload-vs-outcome scatter (filtered with the 0–10,000 kg slider and the site dropdown) shows successful landings across the full payload range — payload mass alone does not determine the outcome. Colour: green = successful landing, red = failed landing.

Predictive Analysis — Machine Learning

Features: one-hot encoded Orbit, LaunchSite, LandingPad, Serial plus FlightNumber, PayloadMass, Flights, GridFins, Reused, Legs, Block, ReusedCount (80 columns), standardized. **Split:** 72 train / 18 test records (80/20, random_state = 2).

Test accuracy by model



Confusion matrix — Decision Tree (test set)

	Predicted: fail (0)	Predicted: land (1)
Actual: fail (0)	**5**	1
Actual: land (1)	0	**12**

5 true negatives, 1 false positive, 0 false negatives, 12 true positives — 17 of 18 correct (94.44%). The tree never missed a successful landing.

Conclusion: the **Decision Tree is the best model (94.44% test accuracy)**, ahead of Logistic Regression (83.33%) and SVM with a sigmoid kernel (83.33%); KNN trails at 66.67%. Landing success can be predicted from launch data with high confidence.

Conclusion

- **Landing success is predictable.** A Decision Tree predicts first-stage landing success with 94.44% test accuracy, so the cost of a Falcon 9 launch can be estimated before flight.
- **Experience matters most.** Success rose from 0% in 2010–2013 to 90.0% in 2019 as SpaceX gained flight experience.
- **Site and orbit matter.** KSC LC 39A is the most reliable site (77.3%); VLEO (85.7%) and SSO (100%) are the most reliable orbits, GTO the hardest (51.9%).
- **Payload is not the deciding factor.** Landings succeed across the full 350–15,600 kg payload range.
- **Reuse drives the cost advantage:** recovering the first stage is what makes a $62M Falcon 9 launch possible.

Project repository (code, notebooks, data and this report):
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Appendix: the full analysis notebook, datasets (dataset_part_1.csv, dataset_part_2.csv) and additional charts are in the GitHub repository. Report prepared November 2026.